A Practical Guide to Time series forecasting

TOPIC -- TIME SERIES FORECASTING

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Time-varying autoregressive (TVAR) models Time-varying autoregressive (TVAR) models are especially useful for analyzing non-stationary time series in which the underlying dynamics evolve over time in complex ways, not limited to classical forms of variation such as trends or seasonal patterns. Unlike classical AR models with fixed parameters, TVAR models allow the autoregressive coefficients to vary as functions of time, typically represented through basis function expansions whose form and complexity are determined by the user, allowing for highly flexible modeling of time-varying behavior, which enables them to capture specific non-stationary patterns exhibited by the data. This flexibility makes them well suited to modeling structural changes, regime shifts, or gradual evolutions in a system's behavior. TVAR time-series models are widely applied in fields such as signal processing, economics, finance, reliability and condition monitoring, telecommunications, neuroscience, climate sciences, and hydrology, where time series often display dynamics that traditional linear models cannot adequately represent. Estimation of TVAR models typically involves methods such as kernel smoothing, recursive least squares, or Kalman filtering.

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Additionally, time series analysis techniques may be divided into parametric and non-parametric methods. The parametric approaches assume that the underlying stationary stochastic process has a certain structure which can be described using a small number of parameters (for example, using an autoregressive or moving-average model). In these approaches, the task is to estimate the parameters of the model that describes the stochastic process. By contrast, non-parametric approaches explicitly estimate the covariance or the spectrum of the process without assuming that the process has any particular structure. Methods of time series analysis may also be divided into linear and non-linear, and univariate and multivariate.

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In mathematics, a time series is a sequence of data points indexed, listed, or graphed in chronological order. Most commonly, a time series consists of observations recorded at successive equally spaced points in time. Thus, it represents a form of discrete-time data. A time series may describe measurements collected over seconds, days, years, or even centuries. Common examples include heights of ocean tides, counts of sunspots, daily temperature readings, and the closing values of stock market indices such as the Dow Jones Industrial Average. A time series is often visualized using a run chart (a type of temporal line chart), which helps identify patterns such as trends, seasonal effects, and irregular fluctuations. Time series are widely used in statistics, actuarial science, signal processing, pattern recognition, econometrics, mathematical finance, weather forecasting, earthquake prediction, electroencephalography, control engineering, astronomy, communications engineering, and many other areas of applied science and engineering that involve temporal measurements. Time series analysis comprises methods for analyzing time series data in order to extract meaningful statistics and other characteristics of the data. Time series forecasting is the use of a model to predict future values based on previously observed values. Generally, time series data is modeled as a stochastic process. While regression analysis is often employed in such a way as to test relationships between one or more different time series, this type of analysis is not usually called "time series analysis", which refers in particular to relationships between different points in time within a single series. Time series data have a natural temporal ordering. This makes time series analysis distinct from cross-sectional studies, in which there is no natural ordering of the observations (e.g. explaining people's wages by reference to their respective education levels, where the individuals' data could be entered in any order). Time series analysis is also distinct from spatial data analysis where the observations typically relate to geographical locations (e.g. accounting for house prices by the location as well as the intrinsic characteristics of the houses). A stochastic model for a time series will generally reflect the fact that observations close together in time will be more closely related than observations further apart. In addition, time series models will often make use of the natural one-way ordering of time so that values for a given period will be expressed as deriving in some way from past values, rather than from future values (see time reversibility). Time series analysis can be applied to real-valued, continuous data, discrete numeric data, or discrete symbolic data (i.e. sequences of characters, such as letters and words in the English language).

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History Jevons was the first to study times series. In his book he examined a number of weekly financial time series between the years 1825 to 1860 inclusive. These included times series of bankruptcies, currency circulation and discount rates.

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Prediction and forecasting In statistics, prediction is a part of statistical inference. One particular approach to such inference is known as predictive inference, but the prediction can be undertaken within any of the several approaches to statistical inference. Indeed, one description of statistics is that it provides a means of transferring knowledge about a sample of a population to the whole population, and to other related populations, which is not necessarily the same as prediction over time. When information is transferred across time, often to specific points in time, the process is known as forecasting.

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Consideration of the autocorrelation function and the spectral density function (also cross-correlation functions and cross-spectral density functions) Scaled cross- and auto-correlation functions to remove contributions of slow components Performing a Fourier transform to investigate the series in the frequency domain Performing a clustering analysis Discrete, continuous or mixed spectra of time series, depending on whether the time series contains a (generalized) harmonic signal or not Use of a filter to remove unwanted noise Principal component analysis (or empirical orthogonal function analysis) Singular spectrum analysis "Structural" models: General state space models Unobserved components models Machine learning Artificial neural networks Support vector machine Fuzzy logic Gaussian process Genetic programming Gene expression programming Hidden Markov model Multi expression programming Queueing theory analysis Control chart Shewhart individuals control chart CUSUM chart EWMA chart Detrended fluctuation analysis Nonlinear mixed-effects modeling Dynamic time warping Dynamic Bayesian network Time-frequency analysis techniques: Fast Fourier transform Continuous wavelet transform Short-time Fourier transform Chirplet transform Fractional Fourier transform Chaotic analysis Correlation dimension Recurrence plots Recurrence quantification analysis Lyapunov exponents Entropy encoding